Padding

Adding one more row and column on top and bottom of the n layer is called padding.
That additional layer will have the values of 0

4×4 n p 2 f 3

Here the output layer = $n + 2p - f + 1$

$= 4 + 4 - 3 + 1$

$= 6 \times 6$

"valid convolution" "Same convolution"

↳ Input and the output has the same number of layer

$$\text{padding layer} = f - 1/2$$

Stride :-

w/o stride $= n + 2p - f + 1$

with stride $= \left\lfloor \frac{n + 2p - f}{s} + 1 \right\rfloor$

f=3 S=2
n=7

2	3	7	4	6	2	9
6	6	9	8	7	4	3
3	4	8	3	8	9	7
7	8	3	6	6	3	4
4	2	1	8	3	4	6
3	2	4	1	9	8	3
0	1	3	9	2	1	4

$$\text{Output} = \left\lfloor \frac{n+2p-f}{s} + 1 \right\rfloor$$

$$= \left\lfloor \frac{7+0-3}{2} + 1 \right\rfloor = 4/2 + 1 = 3$$

Output layer

91	100	88
69	91	117
44	72	74

3 x 2
1 x 6
-1 x 3
6+6-3
= 9
= 91

4 x 3
0 x 6
0 x 4
12
+12 + 70

4 x 7
2 x 9
3 x 8
28+18+24

3 1 4
1 0 4
-1 0 2
3

24
3
-1
12
0
0
32
12
9

18 25
7 42
-8 21
-8 88
0
0
36
6
21

3 x 7
1 x 9
-1 x 8
21+9-8
22+16+62
100

4 x 4
0 x 8
0 x 3
16

4 x 6
2 x 7
3 x 8
24+14+24

9 16
7 16
-4 -4
16 41
0 69
0
32
6
3

24	36	28
6	0	8
-3	0	18
27	36	54

12	8	4
3	0	8
0	0	9

79

8	32	12
4	0	18
-3	0	6

72

9	16	24
9	0	6
-2	0	12

74

29
36
54
119

layer 1

1	2	1	1
2	1	2	-1
-1	2	1	1
1	1	-1	2

1	-1
0	1

0	-1	0	1
1	0	1	0
1	1	-1	-1
2	0	0	1

0	0
1	-1

-1	-2	1	0
1	1	0	-1
-2	1	0	-1
1	1	0	-1

1	1
0	1

output = layer

-1	01	0	
6	3	2	
0	1	-1	

First row First colum.

$1-2+0+1+0+0+1+0+(-1)-2+0+1 =$

$4-5 = -1$

$2-1+0+2+0+0+0-1-2+1+0+0 =$

[illegible] $= 0$

$1-1+0-1+0+0+1+0+1+0+0-1 = 0$

$2-1+0+2+0+0+1-1+1+1+0+1$

$8-2 = 6$

$1-2+0+1+0+0+1+1+1+0+0+0 = 3$

$2+1+0+1+0+0-1+1+0-1+0-1 = 2$

$-1-2+0+1+0+0+2+0$

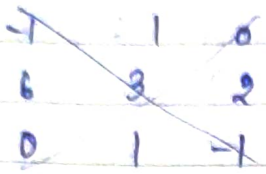
$-2+1+0+1 = 0$

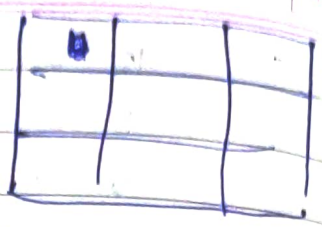
$2-1+0-1+0+0+0+0+0+1+0$

$+0+0$

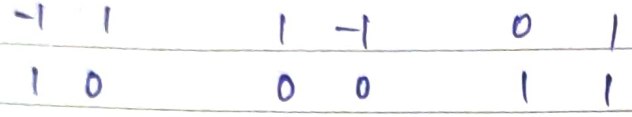
~~$1-1+0+2+0+0+0-1+0-1$~~

$+0-1 = -1$





Multiple filters



$1+1+6+0 \quad -1-1+0+0+0+0+3+2 = 11$

10/08/2026 Lecture 15

One Image → 6x6x3 | One filter → 3x3x3 | output layer → 4x4 | with this add bias term [4x4]+b1 | Apply non Linearity

Image → filter → output layer

Convolution.

1) Convolution
2) Add the bias term
3) Apply the non-linearity.

10 filters ⇒ 3x3x3 in one layer of a neural network how many parameters that layer have?

27 parameters
+ 1 bias
= 28 parameters
⇓
10 filters
⇓
28 x 10 = 280 features.